

Lecture 13 – Eigenvalues

Concepts:

- Eigenvalues
- Matrix determinant
- Complex numbers

Matrix Algebra & Conventions

A - matrix (bold)

$\vec{w}$ - vector (italicized)

A_{ij} or w_i - scalars

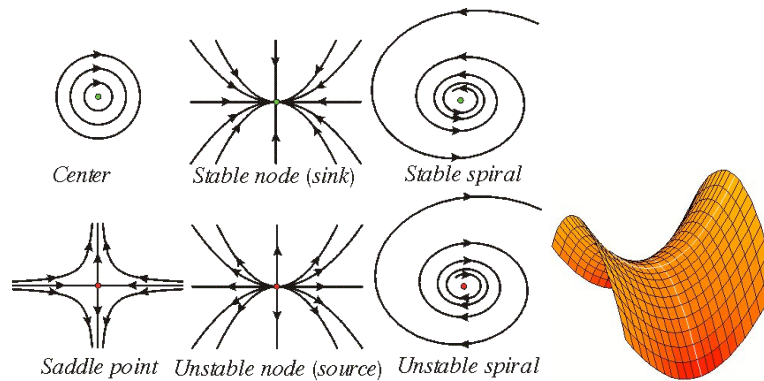
$\begin{bmatrix} a \\ b \end{bmatrix}$

- column vector

$\begin{bmatrix} a & b \end{bmatrix}$ - row vector

Why do eigenvalues indicate stability properties?

Eigenvalues (λ_i)	Interpretation
$Re(\lambda_i) < 0$ for all i	Stable node
$Re(\lambda_i) < 0$ for some i	Saddle point
$Re(\lambda_i) > 0$ for all i	Unstable node
$Re(\lambda_i) = 0$ for all i	Neutrally stable
$Im(\lambda_i) = 0$ for all i	No oscillations
$Im(\lambda_i) \neq 0$ for some i	Oscillations



Recall, for perturbations x and y to species u_1 and u_2 :

$$\frac{dx}{dt} = f_1(u_1^* + x, u_2^* + y) \approx \underbrace{\frac{\partial f_1}{\partial u_1} \Big|_{u_1^*, u_2^*}}_{A_{11}} \cdot x + \underbrace{\frac{\partial f_1}{\partial u_2} \Big|_{u_1^*, u_2^*}}_{A_{12}} \cdot y$$

and

$$\frac{dy}{dt} = f_2(u_1^* + x, u_2^* + y) \approx \underbrace{\frac{\partial f_2}{\partial u_1} \Big|_{u_1^*, u_2^*}}_{A_{21}} \cdot x + \underbrace{\frac{\partial f_2}{\partial u_2} \Big|_{u_1^*, u_2^*}}_{A_{22}} \cdot y$$

Therefore, if x and y are small, can approximate dynamics as

$$\begin{aligned} \frac{dx}{dt} &= A_{11} \cdot x + A_{12} \cdot y \\ \frac{dy}{dt} &= A_{21} \cdot x + A_{22} \cdot y \end{aligned}$$

In matrix form:

$$\underbrace{\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix}}_{\frac{d\vec{n}}{dt}} = \underbrace{\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}}_{\mathbf{A}} \cdot \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\vec{n}}$$

or equivalently

$$\frac{d\vec{n}}{dt} = \mathbf{A} \cdot \vec{n}$$

Side note:

Matrix addition:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$$

Matrix multiplication:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a \cdot e + b \cdot g & a \cdot f + b \cdot h \\ c \cdot e + d \cdot g & c \cdot f + d \cdot h \end{bmatrix}$$

In **R** use **A % * % B** for matrix multiplication (i.e. dot product).

The default is element-wise multiplication (Hadamard product).

Recall integration of single-spp exponential growth model:

$$\frac{dN}{dt} = rN \quad \Rightarrow \quad N(t) = \int_0^t \frac{dN}{dt} dt = N(0) e^{rt}$$

$r < 0 \Rightarrow$ decay.

$r > 0 \Rightarrow$ increase.

Represent dynamics of perturbation is same way.

For i^{th} perturbation:

$$\frac{dn_i}{dt} = A_{i \cdot} \cdot n_i \quad \Rightarrow \quad n_i(t) = n_i(0) e^{A_{i \cdot} t}$$

... or for both perturbations:

$$\vec{n}_t = \vec{n}_0 e^{\mathbf{A}t}$$

But $e^{\mathbf{A}}$ is e^{matrix} for which usual definition of exponential function doesn't work!

How do we make it work?

What if we could replace matrix **A** with some constants – call it/them λ .

That would require that:

$$\boxed{\mathbf{A}\vec{w} = \lambda\vec{w}}$$

Vector $\vec{w}$ the **right eigenvector** of **A**.

$\vec{w}$ is a column vector.

Side-note (we won't actually use this fact)

The **left eigenvector** comes from:

$$\vec{v}\mathbf{A} = \vec{v}\lambda$$

$\vec{v}$ is a row vector.

What is an eigenvalue?

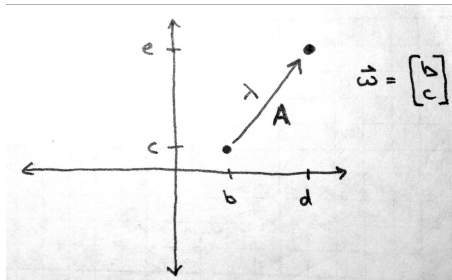
Can think of any square matrix $\mathbf{A}$ and its associated eigenvalues λ as equivalent transformation operations, ...like a multiplication by a 'slope' or 'direction'.

For example, if we let $\mathbf{A} = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$ and $\vec{w} = \begin{bmatrix} b \\ c \end{bmatrix}$.

Then:

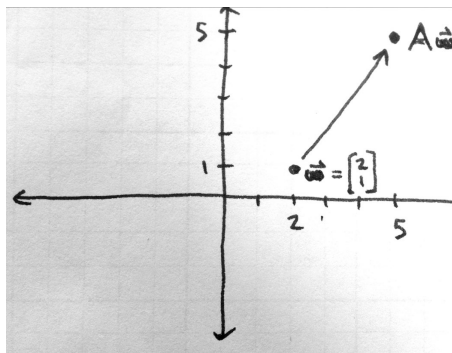
$$\mathbf{A}\vec{w} = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \cdot \begin{bmatrix} b \\ c \end{bmatrix} = \begin{bmatrix} a_1b + a_2c \\ a_3b + a_4c \end{bmatrix} = \begin{bmatrix} d \\ e \end{bmatrix}$$

$$\lambda\vec{w} = \lambda \begin{bmatrix} b \\ c \end{bmatrix} = \begin{bmatrix} \lambda b \\ \lambda c \end{bmatrix} = \begin{bmatrix} d \\ e \end{bmatrix}$$

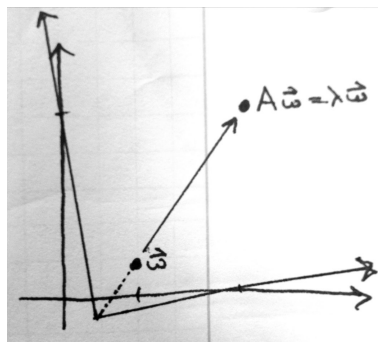


Numerical example:

$$\text{Start at } \vec{w} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ and } \mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4+1 \\ 2+3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$



We can do the same thing with λ by defining a new coordinate system:



How do we determine what λ is?

Need to solve:

$$\mathbf{A}\vec{w} = \lambda\vec{w} \quad \Rightarrow \quad \mathbf{A}\vec{w} - \lambda\vec{w} = 0$$

First we have to write λ in matrix form using **Identity matrix**:

$$\lambda\mathbf{I} = \lambda \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

Thus:

$$\begin{aligned} \mathbf{A}\vec{w} - \lambda\vec{w} &= 0 \\ (\mathbf{A} - \lambda\mathbf{I})\vec{w} &= 0 \end{aligned}$$

Thus the solution is either

$$\vec{w} = 0 \quad \text{or} \quad (\mathbf{A} - \lambda\mathbf{I}) = 0$$

Not interested in $\vec{w} = 0$, but solving $(\mathbf{A} - \lambda\mathbf{I}) = 0$ isn't any easier than what we started with!

However,

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

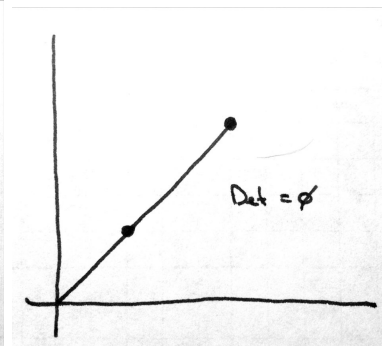
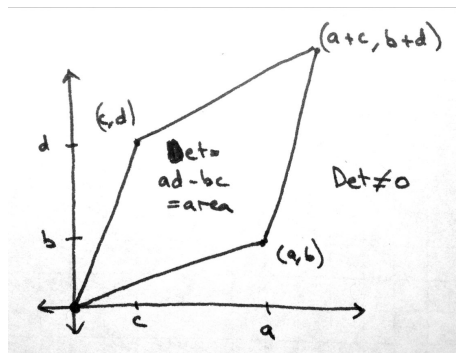
gives us a way to solve it.

What is a matrix determinant?

$$\det \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = ad - bc$$

$$\det \left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \right) = aei + bfg + cdh - ceg - fha - ibd$$

Graphical interpretation of the determinant is as a *volume* (or *area* for 2x2 matrices):



$$\det \left(\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \right) = 2 \cdot 3 - 1 \cdot 1 = 5$$

$$\det \left(\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \right) = 2 \cdot 2 - 4 \cdot 1 = 0$$

Back to solving for eigenvalues

If λ is just a scaling factor, then the volume of $\mathbf{A} - \lambda\mathbf{I}$ must equal 0.

$$\begin{aligned} 0 &= \det(\mathbf{A} - \lambda\mathbf{I}) \\ &= \det \left(\begin{bmatrix} A_{11} - \lambda & A_{12} \\ A_{21} & A_{22} - \lambda \end{bmatrix} \right) \\ &= (A_{11} - \lambda)(A_{22} - \lambda) - A_{12}A_{21} \\ &= \underbrace{\lambda^2 - (A_{11} + A_{22})\lambda + A_{11}A_{22} - A_{12}A_{21}}_{\text{Characteristic equation}} \end{aligned}$$

$\Rightarrow$ Solve for λ by solving the **characteristic equation**.

Solve for λ using quadratic formula:

$$ax^2 + bx + c = 0 \quad \Rightarrow \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

By defining... *(yes, this is confusing, but it's how it's done in the literature)*

$$\lambda^2 - \underbrace{(A_{11} + A_{22})}_{A_1} \lambda + \underbrace{A_{11}A_{22} - A_{12}A_{21}}_{A_2}$$

That is: $A_1 = A_{11} + A_{22}$ and $A_2 = A_{11}A_{22} - A_{12}A_{21}$

...we have $a = 1$, $b = -A_1$ and $c = A_2$.

$$1 \cdot \lambda^2 + (-A_1)\lambda + A_2 = 0 \quad \Rightarrow \quad \begin{cases} \lambda_1 = \frac{A_1 + \sqrt{(-A_1)^2 - 4A_2}}{2} \\ \lambda_2 = \frac{A_1 - \sqrt{(-A_1)^2 - 4A_2}}{2} \end{cases}$$

The two solutions to λ are called a **complex conjugate pair**, or **complex conjugate roots** (=soltns.).

Compare to single-sp. model

For 1-sp. model we used the sign of the slope $\frac{df(N)}{dN}$ to infer stability:

$$\frac{df(N)}{dN} < 0 \quad \Rightarrow \quad \text{stable}$$

Now we're saying that it's the eigenvalues of $\mathbf{A}$ that matter, not the slopes themselves.

But remember that the eigenvalues reflect the partial derivative slopes $\frac{\partial f_i(\vec{N})}{\partial N_j}$.

They reflect the same transformation of the perturbation.

In fact, for a 1-sp. model they are exactly the same:

$$\mathbf{A}\vec{w} = \lambda\vec{w} \quad \Leftrightarrow \quad \frac{df(N)}{dN} \cdot w = \lambda \cdot w \quad \Rightarrow \quad \frac{df(N)}{dN} = \lambda$$

How do Complex numbers arise?

$$\lambda = \frac{A_1 \pm \sqrt{(-A_1)^2 - 4A_2}}{2}$$

When $(-A_1)^2 < 4A_2 \Rightarrow \frac{1}{2}(A_1 \pm \sqrt{\text{negative}})$

But how do we take the $\sqrt{\quad}$ of a negative number?

$\Rightarrow$ Imaginary unit i .

Define $i^2 = -1$

$$\begin{aligned} \sqrt{-\#} &= \sqrt{-1 \cdot \#} = \sqrt{i^2 \#} = \sqrt{i^2} \sqrt{\#} = i\sqrt{\#} \\ \text{e.g., } \sqrt{-4} &= \sqrt{-1 \cdot 4} = \sqrt{i^2 4} = i\sqrt{4} = i2 \end{aligned}$$

$$\lambda = \frac{1}{2}A_1 \pm \frac{1}{2}\sqrt{(-A_1)^2 - 4A_2} i$$

There is nothing 'imaginary' about imaginary numbers! They're as 'real' as negative numbers!

Real numbers are just counts or fractional counts

(= easy to think about as an amount of something).

Can you imagine how much an amount of -10 is? No!

Negative counts don't exist. They're a mathematical convenience to subtract amounts.

Imaginary numbers exist as much as negative numbers do!

They're a mathematical convenience to allow taking roots of negative numbers.

In order to solve... we invented...

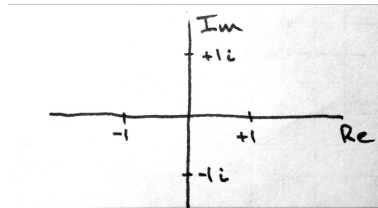
$$\begin{array}{lll}
 x - 5 = 0 & \Rightarrow & \text{Integers} \\
 x + 5 = 0 & \Rightarrow & \text{Negative integers}
 \end{array}
 \left. \vphantom{\begin{array}{l} x - 5 = 0 \\ x + 5 = 0 \end{array}} \right\} \text{ (Real numbers)}$$

$$2x = 1 \quad \Rightarrow \quad \text{Rational numbers (fractions, quotients of integers)}$$

$$x^2 = 2 \quad \Rightarrow \quad \text{Irrational numbers (can't be expressed as fractions, e.g., } \sqrt{2}, \pi, e)$$

$$x^2 = -1 \quad \Rightarrow \quad \text{Imaginary numbers.}$$

Think about imaginary numbers as representing a different number dimension:



Why do imaginary parts indicate oscillations?

Remember that λ represents dynamics of the perturbation:

$$\vec{n}_t = \vec{n}_0 e^{\lambda t}$$

With a complex part:

$$\vec{n}_t = \vec{n}_0 e^{(a \pm bi)t} = e^{at} \cdot e^{\pm ibt}$$

From the theory of complex numbers:

$$e^{i\Theta} = \cos \Theta + i \sin \Theta$$

Multiplying by i is equivalent to rotating by -90° .

Repeat 4 times $\implies$ full rotation (circle).

Rotation viewed from either side $\implies$ sine or cosine.

Thinking of bt as Θ gives:

$$e^{\pm ibt} = \cos(bt) \pm i \sin(bt)$$

Thus:

$$e^{(a \pm bi)t} = e^{at} \cdot [\cos(bt) \pm i \sin(bt)]$$

Therefore, a controls stability

$$\begin{array}{ll}
 a < 0 \Rightarrow & \text{oscillations will dampen} \\
 a > 0 \Rightarrow & \text{oscillations will amplify} \\
 a = 0 \Rightarrow & \text{neutral stability with cycles}
 \end{array}$$

And b controls the *frequency* of the oscillations.

n-spp. stability

Note that we can determine the eigenvalues of an arbitrary-sized square matrix (not just 2-by-2).

We can therefore use eigenvalues to evaluate the local stability of any system of equations for any number of interacting species!